CPE 381 Classwork 12: Fourier Transform

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: October 14, 2026 (In-Class)

1. Fourier transform of a signal $x(t)$ can be calculated as

$$X(\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t}dt$$

Based on the above formula calculate the Fourier transform of the following: **(10 Points)**

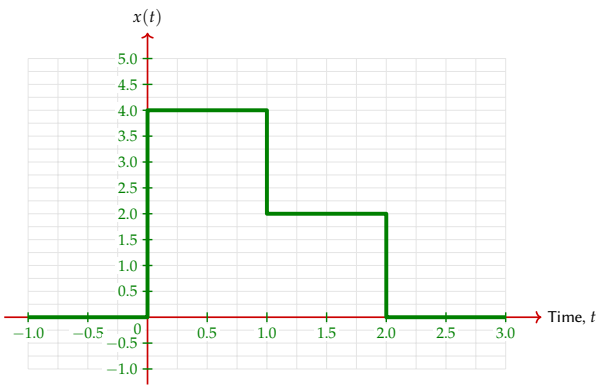


Figure 1: Staircase signal $x(t) = 4\,[u(t) - u(t-1)] + 2\,[u(t-1) - u(t-2)]$.

Answer

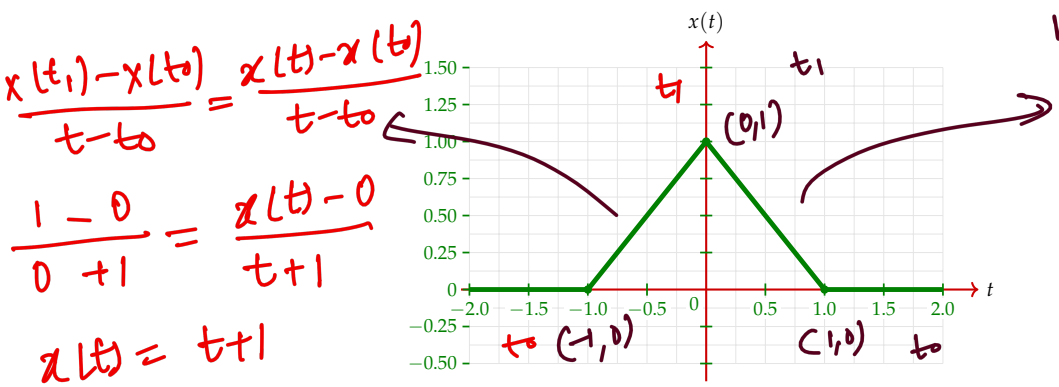
$$X(\Omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\Omega t}\, dt$$

$$= \int_{0}^{1} 4 \cdot e^{-j\Omega t}\, dt + \int_{1}^{2} 2 \cdot e^{-j\Omega t}\, dt$$

$$= 4 \left[-\frac{1}{j\Omega} e^{-j\Omega t} \right]_0^1 + 2 \left[-\frac{1}{j\Omega} e^{-j\Omega t} \right]_1^2$$

$$= 4\left[\frac{e^{-j\Omega}-1}{-j\Omega}\right] + 2\left[\frac{e^{-2j\Omega}-e^{-j\Omega}}{-j\Omega}\right]$$

2. Find the fourier transform of the triangle function show below: **(10 Points)**



$\frac{x(t_1)-x(t_0)}{t-t_0} = \frac{x(t)-x(t_0)}{t-t_0}$

$\frac{1-0}{0+1} = \frac{x(t)-0}{t+1}$

$x(t) = t+1$

We need to find the line equations first

$\frac{x(t_1)-x(t_0)}{t_1-t_0} = \frac{x(t)-x(t_0)}{t-t_0}$

$= \frac{1-0}{0-1} = \frac{x(t)-0}{t-1}$

$\Rightarrow -1 = \frac{x(t)}{t-1}$

$\Rightarrow x(t) = 1-t$

Figure 2: Triangle Fnction

Answer

So $X(\Omega) = \int_{-1}^{0}(t+1)e^{-j\Omega t}dt = \int_{0}^{1}(1-t)e^{-j\Omega t}dt$

$= \int_{-1}^{0} t\cdot e^{-j\Omega t}dt + \int_{-1}^{0} e^{-j\Omega t}dt + \int_{0}^{1} e^{-j\Omega t}dt - \int_{0}^{1} te^{-j\Omega t}dt$

$= \left[\frac{-t}{j\Omega}e^{-j\Omega t} - \frac{1}{(j\Omega)^2}e^{-j\Omega t}\right]_{-1}^{0} + \left[-\frac{1}{j\Omega}e^{-j\Omega t}\right]_{-1}^{0} + \left[-\frac{1}{j\Omega}e^{-j\Omega t}\right]_{0}^{1} + \left[\frac{-t}{j\Omega}e^{-j\Omega t} - \frac{1}{(j\Omega)^2}e^{-j\Omega t}\right]_{0}^{1}$

$\int udv = uv - \int vdu$

$u = t \quad du = dt$

$dv = e^{-j\Omega t}$

$v = -\frac{1}{j\Omega}e^{-j\Omega t}$

$\int te^{-j\Omega t} = -\frac{t}{j\Omega}e^{-j\Omega t} - \int -\frac{1}{j\Omega}e^{-j\Omega t}dt$

$\int te^{-j\Omega t}dt = -\frac{t}{j\Omega}e^{-j\Omega t} - \frac{1}{(j\Omega)^2}e^{-j\Omega t}$

$$= \left[0 - \frac{1}{(j\Omega)^2} e^{-j\Omega \cdot 0}\right] - \left[\frac{e^{j\Omega}}{j\Omega} - \frac{1}{(j\Omega)^2} e^{j\Omega}\right]$$

$$+ \left[-\frac{1}{j\Omega} + \frac{1}{j\Omega} e^{j\Omega}\right]$$

$$+ \left[-\frac{1}{j\Omega} e^{j\Omega} + \frac{1}{j\Omega}\right] + \left[-\frac{1}{j\Omega} e^{-j\Omega} - \frac{1}{(j\Omega)^2} e^{-j\Omega}\right]$$

$$- \left[0 - \frac{1}{(j\Omega)^2}\right]$$

$$= -\frac{1}{(j\Omega)^2} - \frac{e^{j\Omega}}{j\Omega} + \frac{1}{(j\Omega)^2} e^{j\Omega} - \frac{1}{j\Omega} + \frac{1}{j\Omega} e^{j\Omega}$$

$$- \frac{1}{j\Omega} e^{-j\Omega} + \frac{1}{j\Omega} - \frac{1}{j\Omega} e^{j\Omega} - \frac{1}{(j\Omega)^2} e^{-j\Omega}$$

$$= \frac{1}{(j\Omega)^2}\left[e^{j\Omega} - e^{-j\Omega}\right] - \frac{2}{j\Omega} e^{j\Omega} + \frac{1}{(j\Omega)^2}$$